Lesson 5.1 - Introduction to Probability

> **Video:** Introduction to Probability, Basic Overview - Sample Space, & Tree Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Probability of an event

$$P(A) = \frac{\text{number of favorable outcomes (outcomes in } A)}{\text{total number of possible outcomes}}$$

This counting formula works when all outcomes are **equally likely**.

Range of values

$$0 \leq P(A) \leq 1$$

$P(A)$	Meaning
0	Impossible, it never happens
1	Certain, it always happens (100%)
0.3	30% chance: about 3 times in 10 tries, or 30 in 100

Example: if 20% of a city's residents drive a blue car, then picking 100 people at random should give **about 20** blue-car drivers, and picking 1,000 should give about 200.

Sample space and tree diagrams

The **sample space** S is the set of **all possible outcomes**.

One coin: $S = \{H, T\}$

Two coins: use a tree diagram.

```
        H — HH
    H <
        T — HT
start
        H — TH
    T <
        T — TT
```

$S = \{HH, HT, TH, TT\}$, so 4 outcomes.

Three coins:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, so $2^3 = 8$ outcomes.

> In general, flipping n coins gives 2^n equally likely outcomes.

Example 1 - coins

(a) Two coins: P(at least one head)? Favorable: $\{HH, HT, TH\}$, which is 3 of 4 outcomes.

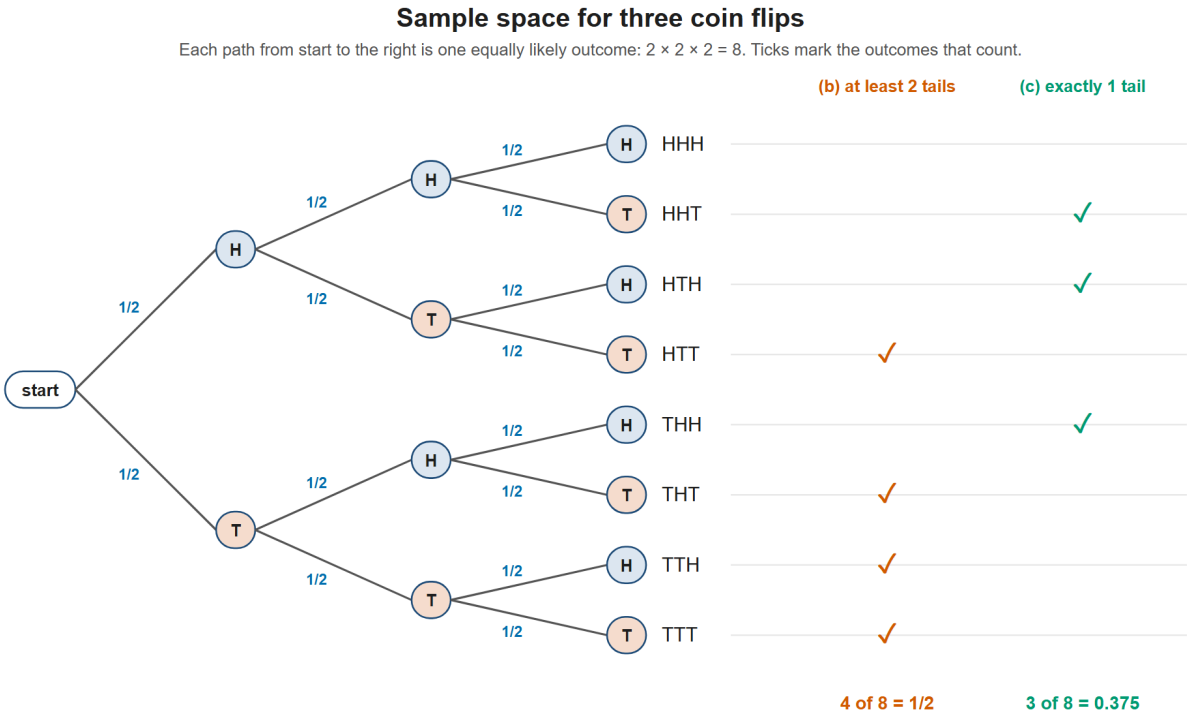
$$P = \frac{3}{4} = 0.75 = 75\%$$

(b) Three coins: P(at least two tails)? Favorable: $\{HTT, THT, TTH, TTT\}$, which is 4 of 8.

$$P = \frac{4}{8} = \frac{1}{2} = 50\%$$

(c) Three coins: P(exactly one tail)? Favorable: $\{HHT, HTH, THH\}$, which is 3 of 8.

$$P = \frac{3}{8} = 0.375 = 37.5\%$$

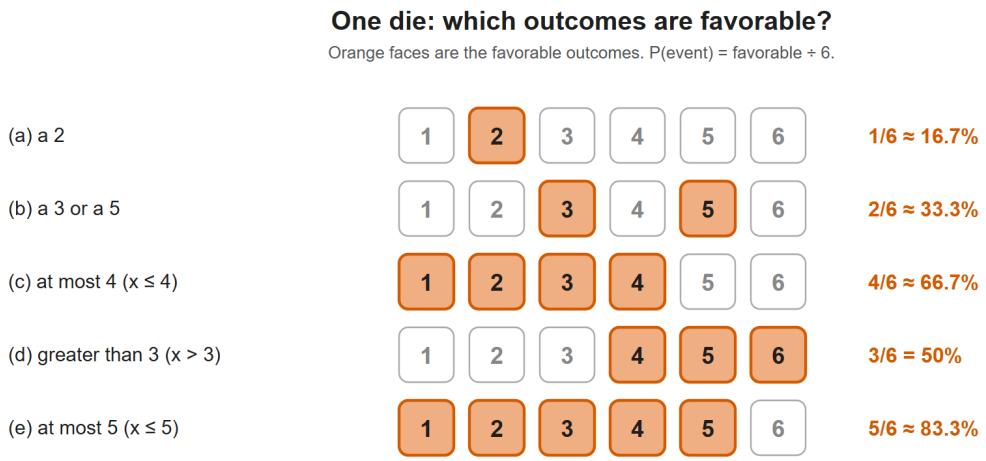


Every path through the tree is one of the 8 equally likely outcomes. Count the ticks: 4 outcomes have at least two tails (4/8 = 1/2), and 3 have exactly one tail (3/8).

Example 2 - one six-sided die

$S = \{1, 2, 3, 4, 5, 6\}$

Event	Favorable outcomes	Probability
(a) a 2	{2}	$\frac{1}{6} \approx 0.167$ (16.7%)
(b) a 3 or a 5	{3, 5}	$\frac{2}{6} = \frac{1}{3} \approx 33.3\%$
(c) **at most** 4 ($x \leq 4$)	{1, 2, 3, 4}	$\frac{4}{6} = \frac{2}{3} \approx 66.7\%$
(d) **greater than** 3 ($x > 3$, so 3 is excluded)	{4, 5, 6}	$\frac{3}{6} = \frac{1}{2} = 50\%$
(e) at most 5 ($x \leq 5$)	{1, 2, 3, 4, 5}	$\frac{5}{6} \approx 83.3\%$



The favorable faces for each die question. Note how "at most 4" includes 4, while "greater than 3" leaves 3 out.

Key takeaways

- $P(A)$ = favorable ÷ total, when the outcomes are equally likely.
- Every probability is between 0 and 1. Multiply by 100 for a percentage.
- List the **sample space**; a **tree diagram** helps for multi-step experiments.
- Read the wording carefully: "at least", "at most", "greater than" (strict), "exactly".

Lesson 5.2 - Probability Formulas

> **Video:** Probability Formulas, Symbols & Notations - Marginal, Joint, & Conditional Probabilities by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

This lesson collects the main probability formulas and symbols in one place. All examples use this setup:

- Sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $A = \{1, 2, 3, 4\}, \quad B = \{6, 7, 8\}, \quad C = \{1, 7, 8, 9\}$

---

1. Marginal probability - a single event

The probability of **one** event, regardless of any others:

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

- $P(A) = \frac{4}{9} \approx 0.444$ (44.4%)
- $P(B) = \frac{3}{9} = \frac{1}{3} \approx 0.333$ (33.3%)
- $P(C) = \frac{4}{9} \approx 0.444$ (44.4%)

---

2. Union and intersection symbols

Symbol	Name	Meaning	Example
$A \cup B$	Union	Everything in A **or** B (or both), listed once	$B \cup C = \{1, 6, 7, 8, 9\}$
$A \cap B$	Intersection	Only what is in **both** A **and** B	$B \cap C = \{7, 8\}$

$A \cap B = \varnothing$ (no shared elements), so A and B are **mutually exclusive**.

---

3. Union probability - $P(A \cup B)$ (addition rule)

$P(A \cup B)$ is read "the probability of A **or** B" (at least one of them happens).

$$\begin{aligned} \text{Words:} \quad & P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \\ \text{Symbols:} \quad & P(A \cup B) = P(A) + P(B) - P(A \cap B) \end{aligned}$$

The subtraction stops the overlap $A \cap B$ from being counted **twice**.

Mutually exclusive events can't happen together, so $A \cap B = \varnothing$ and $P(A \cap B) = 0$:

$$\begin{aligned} \text{Words:} \quad & P(A \text{ or } B) = P(A) + P(B) \\ \text{Symbols:} \quad & P(A \cup B) = P(A) + P(B) \end{aligned}$$

Example - $A \cup B$ (mutually exclusive): $A \cup B = \{1, 2, 3, 4, 6, 7, 8\}$

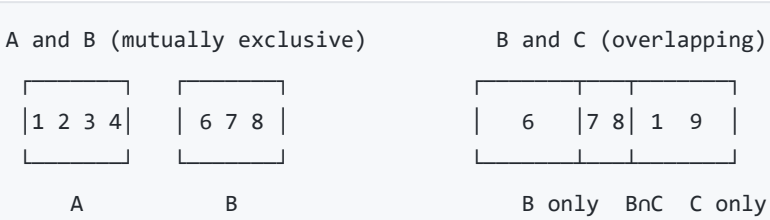
$$P(A \cup B) = \frac{7}{9} \approx 0.778 \quad \left(= \tfrac{4}{9} + \tfrac{3}{9}\right)$$

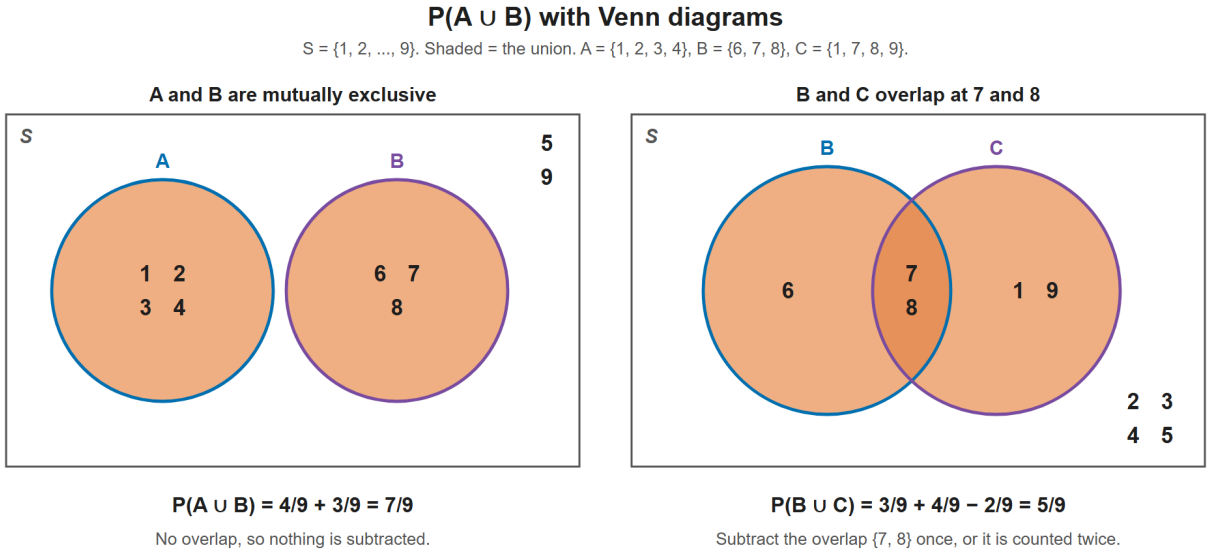
Example - $B \cup C$ (they overlap at 7 and 8): $B \cap C = \{7, 8\}$, so $P(B \cap C) = \frac{2}{9}$

$$P(B \cup C) = P(B) + P(C) - P(B \cap C) = \frac{3}{9} + \frac{4}{9} - \frac{2}{9} = \frac{5}{9}$$

Check by listing: $B \cup C = \{1, 6, 7, 8, 9\}$ has 5 elements → $\frac{5}{9}$ ✓. Without subtracting $\frac{2}{9}$, the 7 and 8 would be counted twice.

Venn diagrams:





In the left diagram the circles don't touch, so $P(A \cup B)$ is just $\frac{4}{9} + \frac{3}{9}$. In the right diagram 7 and 8 sit in both circles. Adding $\frac{3}{9}$ and $\frac{4}{9}$ would count them twice, so $P(B \cap C) = \frac{2}{9}$ is subtracted once.

> **Word cue:** "**or**" → ∪ → **addition** rule. "**and**" → ∩ → **multiplication** rule.

---

4. Joint probability - $P(A \cap B)$ (multiplication rule)

$P(A \cap B)$ is read "the probability of A **and** B" (both happen):

$$\begin{aligned} \text{Words:} \quad & P(A \text{ and } B) = P(A \text{ given } B)\, P(B) = P(B \text{ given } A)\, P(A) \\ \text{Symbols:} \quad & P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A) \end{aligned}$$

Independent events (one does not affect the other): $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$, so

$$\begin{aligned} \text{Words:} \quad & P(A \text{ and } B) = P(A)\, P(B) \\ \text{Symbols:} \quad & P(A \cap B) = P(A)\, P(B) \end{aligned}$$

Independent or dependent?

Situation	Type	Why
Draw a red marble, **put it back**, then draw a blue one	**Independent**	Replacing the marble restores the bag
Draw a red marble, **keep it**, then draw a blue one	**Dependent**	The bag now has one fewer marble, so the second probability changes
Toss a coin twice	**Independent**	The first toss doesn't affect the second

Notation warning: $P(A \cap B)$ vs $P(AB)$

- $P(A \cap B) = P(B \cap A)$: both happen, no order implied.
- Some books write $P(AB)$ for "A **first, then** B" and $P(BA)$ for "B **first, then** A". These are **not always equal**.
- Read the context: are the events **simultaneous**, or does one come **before** the other?

---

5. Conditional probability - "A given B"

$P(A \mid B)$ is the probability of A **given that** B **has already happened**:

$$\text{Words:} \quad P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}, \qquad P(B \text{ given } A) = \frac{P(A \text{ and } B)}{P(A)}$$

$$\text{Symbols:} \quad P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Conditional = joint ÷ marginal.

---

6. Bayes' theorem

Replacing $P(A \cap B)$ with $P(B \mid A)\,P(A)$ from the multiplication rule gives

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

It is the conditional probability formula with the joint probability rewritten.

---

7. Negation (complement) probability

$$\text{Words:} \quad P(\text{not } A) = 1 - P(A)$$

$$\text{Symbols:} \quad P(A') = 1 - P(A)$$

$P(A')$ is read "the probability of **not** A". The complement is also written A^c (the most common form in textbooks), $\bar{A}$ or $\sim A$.

- $P(A') = 1 - \frac{4}{9} = \frac{5}{9}$. Check: S has 5 elements outside A (5, 6, 7, 8, 9) ✓
- If $P(B) = 0.35$, then $P(B') = 1 - 0.35 = 0.65$, a 65% chance that B does not occur.

This works because the total probability of all outcomes is 1.

Formula summary

Type	Words	Symbols
Marginal	$P(A) = \frac{\text{favorable}}{\text{total}}$	(same)
Union (or)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Union, mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B)$
Joint (and)	$P(A \text{ and } B) = P(A \text{ given } B)P(B) = P(B \text{ given } A)P(A)$	$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$
Joint, independent	$P(A \text{ and } B) = P(A)P(B)$	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Bayes	$P(A \text{ given } B) = \frac{P(B \text{ given } A)P(A)}{P(B)}$	$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(A') = 1 - P(A)$

Lesson 5.3 - Probability of Complementary Events

> **Video:** Probability of Complementary Events & Sample Space by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Setup - rolling one six-sided die

- Sample space $S = \{1, 2, 3, 4, 5, 6\}$
- A = numbers **at most 4** = $\{1, 2, 3, 4\}$
- $B = \{3, 4, 5\}$

---

1. $P(A \text{ and } B)$ - intersection

$A \cap B = \{3, 4\}$ (the outcomes in **both** sets)

$$P(A \text{ and } B) = \frac{2}{6} = \frac{1}{3} \approx 0.33$$

2. $P(A \text{ or } B)$ - union

$A \cup B = \{1, 2, 3, 4, 5\}$ (the outcomes in **either** set; only 6 is missing)

$$P(A \text{ or } B) = \frac{5}{6} \approx 0.833 = 83.3\%$$

3. $P(A')$ - complement

The **complement** A' is everything in S that is **not** in A: $A' = \{5, 6\}$

$$P(A') = \frac{2}{6} = \frac{1}{3}$$

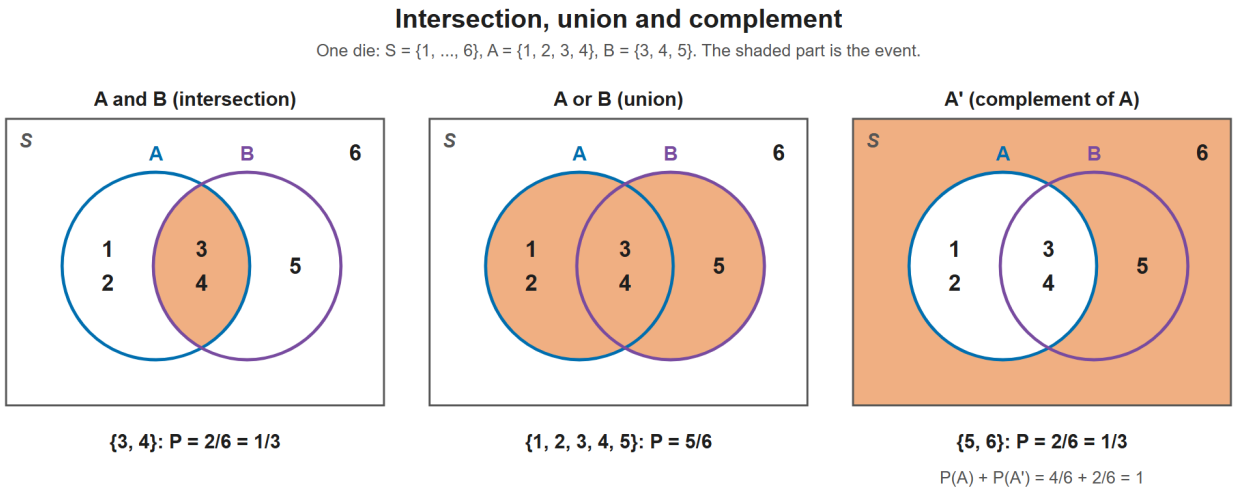
4. $P(A)$ and the complement rule

$$P(A) = \frac{4}{6} = \frac{2}{3}$$

$$P(A) + P(A') = \frac{2}{3} + \frac{1}{3} = 1$$

An event and its complement together cover the whole sample space, so their probabilities add to 1:

$$\boxed{P(A') = 1 - P(A)}$$



The same two events shaded three ways. The complement A' is everything in S outside A, so A and A' together fill the whole box: P(A) + P(A') = 1.

Example - marbles

The probability of drawing a **red** marble from a bag is 35%. What is the probability of drawing a marble that is **not red**?

$$P(\text{not red}) = 1 - 0.35 = 0.65 = 65\%$$

Key takeaways

- Intersection (A and B) = outcomes in **both**. Union (A or B) = outcomes in **at least one**.
- Complement A' = outcomes **not** in A.
- $P(A) + P(A') = 1$, so $P(A') = 1 - P(A)$. This is often the fastest route, especially for "at least one" problems.

Lesson 5.4 - Conditional Probability with Venn Diagrams and Contingency Tables

> **Video:** Conditional Probability With Venn Diagrams & Contingency Tables by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

What it means

$P(A \mid B)$ is the probability that A happens **given that B has already happened**.

Formula:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

Shortcut: ask "*how much of B is also A?*"

$$P(A \mid B) = \frac{\text{count in both } A \text{ and } B}{\text{count in } B}$$

Once B is known to have happened, B becomes the new sample space.

Example 1 - a die

$S = \{1, \ldots, 6\}$, A = odd = $\{1, 3, 5\}$, $B = \{3, 4, 5\}$

Shortcut: of the 3 outcomes in B, two (3 and 5) are in A:

$$P(A \mid B) = \frac{2}{3}$$

Formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3} \checkmark$$

Example 2 - Venn diagram (algebra and chemistry)

A school has **500** students. **150** take algebra (A), **80** take chemistry (C), and **30** take both.

The four regions of the Venn diagram (two overlapping circles inside a box of 500):

Region	Count
Algebra only (A but not C)	150 − 30 = **120**
Both ($A \cap C$, the overlap)	**30**
Chemistry only (C but not A)	80 − 30 = **50**
Neither (outside both circles)	500 − 200 = **300**
Total	**500**

(a) P(taking algebra)

$$P(A) = \frac{150}{500} = \frac{3}{10} = 0.30$$

(b) P(chemistry | algebra): "how much of A is also C?"

$$P(C \mid A) = \frac{30}{150} = \frac{1}{5} = 0.20$$

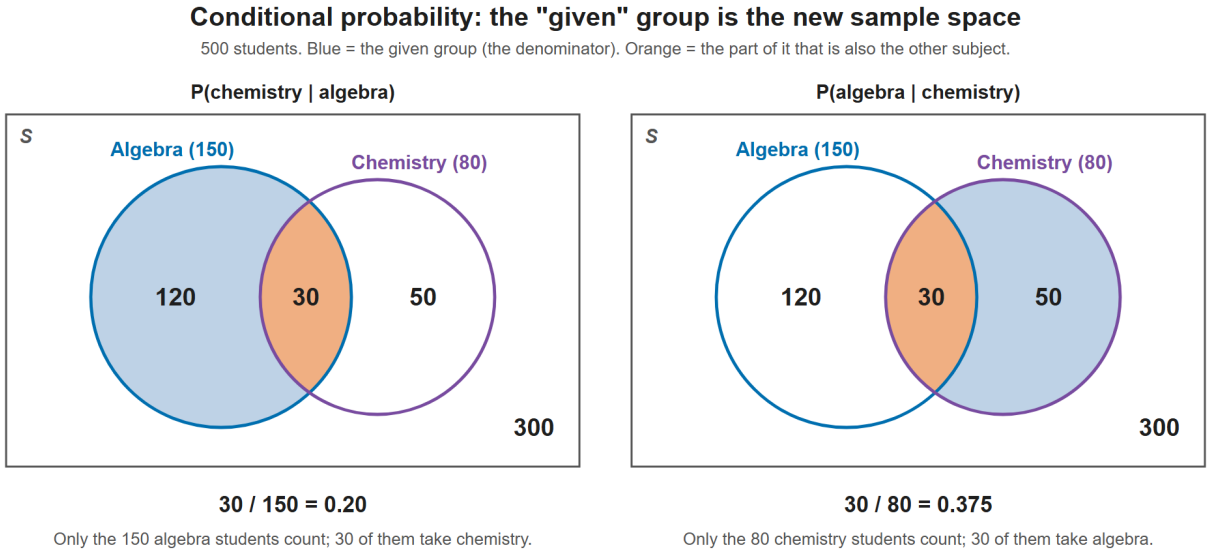
20% of the 150 algebra students also take chemistry: $0.20 \times 150 = 30$ ✓.

Formula check: $\frac{P(C \cap A)}{P(A)} = \frac{30/500}{150/500} = \frac{30}{150}$ ✓

(c) P(algebra | chemistry): "how much of C is also A?"

$$P(A \mid C) = \frac{30}{80} = \frac{3}{8} = 0.375$$

> $P(C \mid A) \neq P(A \mid C)$. The order matters, because the denominator is the "given" group.



The given group shrinks the sample space. P(chemistry | algebra) only looks at the 150 algebra students, and P(algebra | chemistry) only at the 80 chemistry students, so the same overlap of 30 gives two different answers.

Example 3 - contingency table (birds in a zoo)

200 birds: 70 male with brown eyes, 100 female with brown eyes, 20 male with blue eyes, 10 female with blue eyes.

	Blue eyes	Brown eyes	**Total**
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

(a) P(female)

$$\frac{110}{200} = \frac{11}{20} = 0.55$$

(b) P(male *and* brown eyes): a joint probability, not a conditional one

$$\frac{70}{200} = 0.35$$

(c) P(female | brown eyes): restrict to the 170 brown-eyed birds

$$\frac{100}{170} = \frac{10}{17} \approx 0.588$$

So 58.8% of the brown-eyed birds are female.

(d) P(male | blue eyes): restrict to the 30 blue-eyed birds

$$\frac{20}{30} = \frac{2}{3} \approx 0.667$$

(e) P(blue eyes | female): restrict to the 110 female birds

$$\frac{10}{110} = \frac{1}{11} \approx 0.091 = 9.1\%$$

> **Correction:** the video says 99.1% when converting this answer. $1/11 \approx 0.091$, which is **9.1%**.

Joint vs conditional probability in a contingency table

Orange = numerator. Blue outline = denominator. Light blue = the row or column you restrict to.

(b) P(male and brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

70 / 200 = 0.35

joint: divide by the grand total

(c) P(female | brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

100 / 170 ≈ 0.588

given brown: divide by the column total

(e) P(blue | female)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

10 / 110 ≈ 0.091

given female: divide by the row total

The numerator is always one cell. What changes is the denominator: the grand total for a joint probability, or one row or column total for a conditional probability.

Key takeaways

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, or with counts, $\dfrac{n(A \cap B)}{n(B)}$.
- The "given" event becomes the **denominator**, the new reduced sample space.
- In a contingency table, a conditional probability uses **one row or column total** as the denominator. A joint probability uses the **grand total**.
- $P(A \mid B)$ and $P(B \mid A)$ are usually different.

Lesson 5.5 - Probability of Independent and Dependent Events

> **Video:** Probability - Independent and Dependent Events by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Notation: "A then B" as an intersection

The video says "$P(A \text{ then } B)$". Formally this is an **intersection** ("and"), with the order recorded by subscripts:

- A_1 = "A happens on the 1st draw"
- B_2 = "B happens on the 2nd draw"

$$P(A \text{ then } B) = P(A_1 \cap B_2)$$

The symbol $\cap$ means "and". Since $A \cap B = B \cap A$, the order ("then") comes from the subscripts, not from $\cap$.

The multiplication rule (general form)

From the definition of conditional probability, $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$, multiplying both sides by $P(A)$ gives:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2 \mid A_1)$$

This always holds. $P(B_2 \mid A_1)$ reads "the probability of B on the 2nd draw, **given** A on the 1st".

Definitions

- **Independent events:** the outcome of one does **not** change the probability of the other, i.e. $P(B_2 \mid A_1) = P(B_2)$, so the condition drops out:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2)$$

- **Dependent events:** the first outcome **changes** the probability of the second, so the conditional probability must be kept:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2 \mid A_1)$$

Drawing marbles...	Type	Why
with replacement	Independent	The marble goes back, so the bag is unchanged
without replacement	Dependent	The bag has one fewer marble (and maybe one fewer of a color)

Example - a bag of marbles

Red	Blue	Green	Yellow	**Total**
8	7	6	4	**25**

Notation: R_1 = red on draw 1, G_2 = green on draw 2, and so on.

(a) P(red)

$$P(R) = \frac{8}{25} = 0.32 = 32\%$$

(b) Blue, then green, *with* replacement (independent)

$$P(B_1 \cap G_2) = P(B_1) \cdot P(G_2) = \frac{7}{25} \cdot \frac{6}{25} = \frac{42}{625} = 0.0672 = 6.72\%$$

The blue marble is put back, so there are still 25 marbles and 6 green ones on the second draw.

(c) Yellow, then red, *without* replacement (dependent)

$$P(Y_1 \cap R_2) = P(Y_1) \cdot P(R_2 \mid Y_1) = \frac{4}{25} \cdot \frac{8}{24} = \frac{4}{25} \cdot \frac{1}{3} = \frac{4}{75} \approx 0.0533 = 5.33\%$$

After the yellow marble is removed, only **24** marbles are left (still 8 red).

(d) Two blue, *with* replacement

$$P(B_1 \cap B_2) = P(B_1) \cdot P(B_2) = \frac{7}{25} \cdot \frac{7}{25} = \frac{49}{625} = 0.0784 = 7.84\%$$

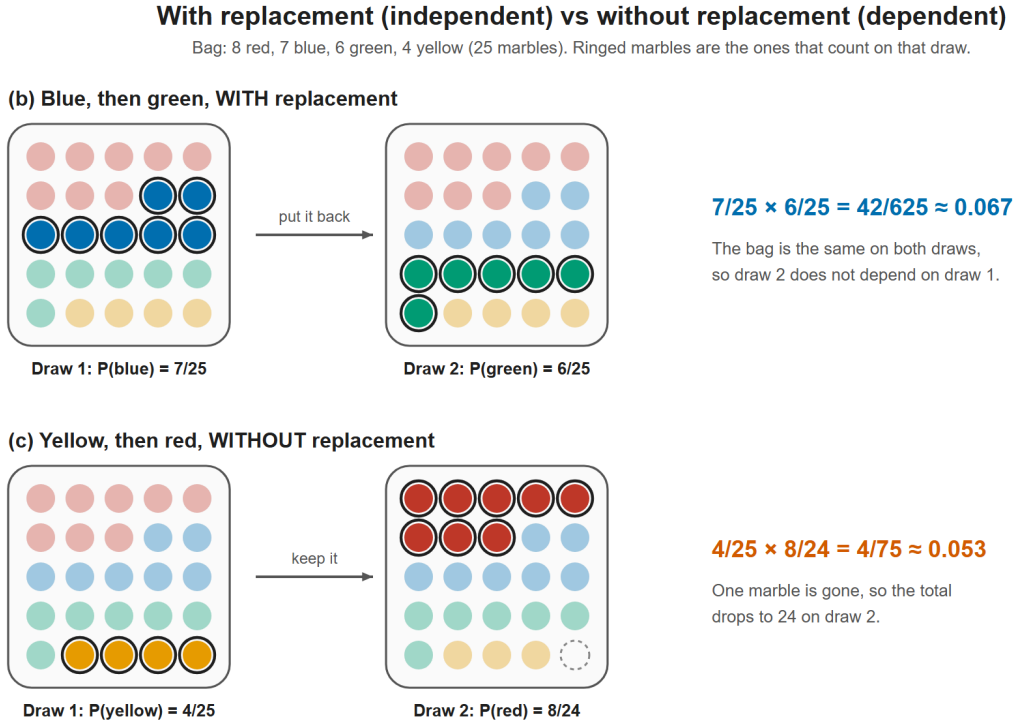
(e) Two green, *without* replacement

$$P(G_1 \cap G_2) = P(G_1) \cdot P(G_2 \mid G_1) = \frac{6}{25} \cdot \frac{5}{24} = \frac{30}{600} = \frac{1}{20} = 0.05 = 5\%$$

After one green marble is removed, there are **5 green** out of **24**.

Why (b) is independent and (c) is dependent

- In (b), $P(G_2 \mid B_1) = \frac{6}{25}$, which is exactly $P(G_2)$ on its own. The first draw had no effect.
- In (c), $P(R_2 \mid Y_1) = \frac{8}{24}$ instead of $P(R_2) = \frac{8}{25}$. The first draw changed it.



With replacement the second bag is identical to the first, so the draws are independent. Without replacement one marble is missing, so the second probability uses 24 marbles.

Key takeaways

- **With replacement → independent:** multiply the unchanged probabilities.
- **Without replacement → dependent:** after each draw, reduce the total and (if needed) the count of the color drawn.
- "First … then …" problems use the **multiplication rule**: $P(A_1 \cap B_2) = P(A_1) \cdot P(B_2 \mid A_1)$.
- For three or more draws it chains: $P(A_1 \cap B_2 \cap C_3) = P(A_1) \cdot P(B_2 \mid A_1) \cdot P(C_3 \mid A_1 \cap B_2)$.

Lesson 5.6 - Probability of Mutually Exclusive Events with Venn Diagrams

> **Video:** Probability of Mutually Exclusive Events With Venn Diagrams by The Organic Chemistry Tutor
> **Chapter:** 5 - Introduction to Probability

Definition

Events A and B are **mutually exclusive** if they **cannot happen at the same time**, meaning they share **no outcomes**:

$$A \cap B = \varnothing \quad \iff \quad P(A \text{ and } B) = 0$$

Venn diagram: mutually exclusive events are drawn as **separate, non-overlapping** circles. Events that are not mutually exclusive are drawn as **overlapping** circles.

Testing for mutual exclusivity

One die, $S = \{1, 2, 3, 4, 5, 6\}$:

- $A = \{1, 2, 3\},\ B = \{5, 6\},\ C = \{3, 4, 5\}$

Pair	Intersection	$P(\text{both})$	Mutually exclusive?
A, B	$\varnothing$	0	**Yes**
A, C	{3}	1/6	No
B, C	{5}	1/6	No

> **Rule:** if $P(A \text{ and } B) = 0$, they are mutually exclusive. If it is anything other than 0, they are not.

The addition rule

General (always works):

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Mutually exclusive (the last term is 0):

$$P(A \text{ or } B) = P(A) + P(B)$$

Example 1 - overlapping events

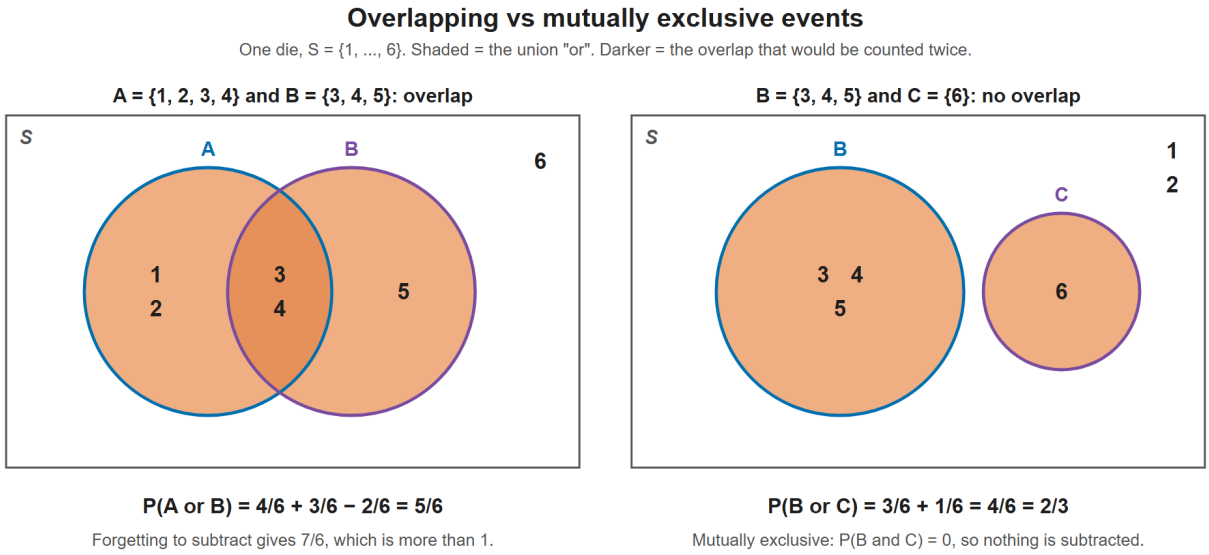
$A = \{1, 2, 3, 4\},\ B = \{3, 4, 5\}$. They share 3 and 4, so they are **not** mutually exclusive.

Venn diagram regions: A only = {1, 2}, both = {3, 4}, B only = {5}, outside both = {6}.

$$P(A \text{ or } B) = \frac{4}{6} + \frac{3}{6} - \frac{2}{6} = \frac{5}{6}$$

Without subtracting $\frac{2}{6}$ you would get $\frac{7}{6}$, which is more than 1 and therefore impossible. The subtraction removes the double-counted 3 and 4.

Check: $A \cup B = \{1, 2, 3, 4, 5\}$, which is 5 of 6 outcomes ✓



Overlapping circles share outcomes, so the overlap must be subtracted once. Separate circles share nothing, so P(A or B) is simply P(A) + P(B).

Example 2 - mutually exclusive events

$B = \{3, 4, 5\},\ C = \{6\}$. They share nothing, so they are mutually exclusive and the circles don't overlap.

$$P(B \text{ or } C) = \frac{3}{6} + \frac{1}{6} - 0 = \frac{4}{6} = \frac{2}{3}$$

Key takeaways

- Mutually exclusive = no shared outcomes = $P(A \cap B) = 0$ = non-overlapping circles.
- Use $P(A) + P(B) - P(A \cap B)$ in general. For mutually exclusive events it simplifies to $P(A) + P(B)$.
- A probability above 1 is a sign that an overlap was counted twice.

Lesson 5.7 - Addition Rule and Multiplication Rule

> **Video:** Multiplication & Addition Rule - Probability - Mutually Exclusive & Independent Events by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Addition rule - "or"

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \text{ and } B)$$

- **Mutually exclusive** events: $P(A \text{ and } B) = 0$, so $P(A \text{ or } B) = P(A) + P(B)$.
- The general form **always** works.

Multiplication rule - "and"

Start from conditional probability:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)} \;\Rightarrow\; P(A \text{ and } B) = P(A \mid B)\,P(B)$$

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)} \;\Rightarrow\; P(A \text{ and } B) = P(B \mid A)\,P(A)$$

Independent events: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$, so

$$P(A \text{ and } B) = P(A)\,P(B)$$

> "A and B" means both happen **together**, so $P(A \text{ and } B) = P(B \text{ and } A)$. This is not the same as "A **then** B", where order can matter.

Tests

Property	Test
Independent	$P(A \mid B) = P(A)$, or equivalently $P(A \text{ and } B) = P(A)P(B)$
Mutually exclusive	$P(A \text{ and } B) = 0$

Example - Sarah's course choices

- $P(A)$ = P(enrolls in algebra) = 0.30
- $P(B)$ = P(enrolls in biology) = 0.70
- $P(A \mid B)$ = P(algebra | biology) = 0.40

(a) P(algebra *and* biology) - multiplication rule

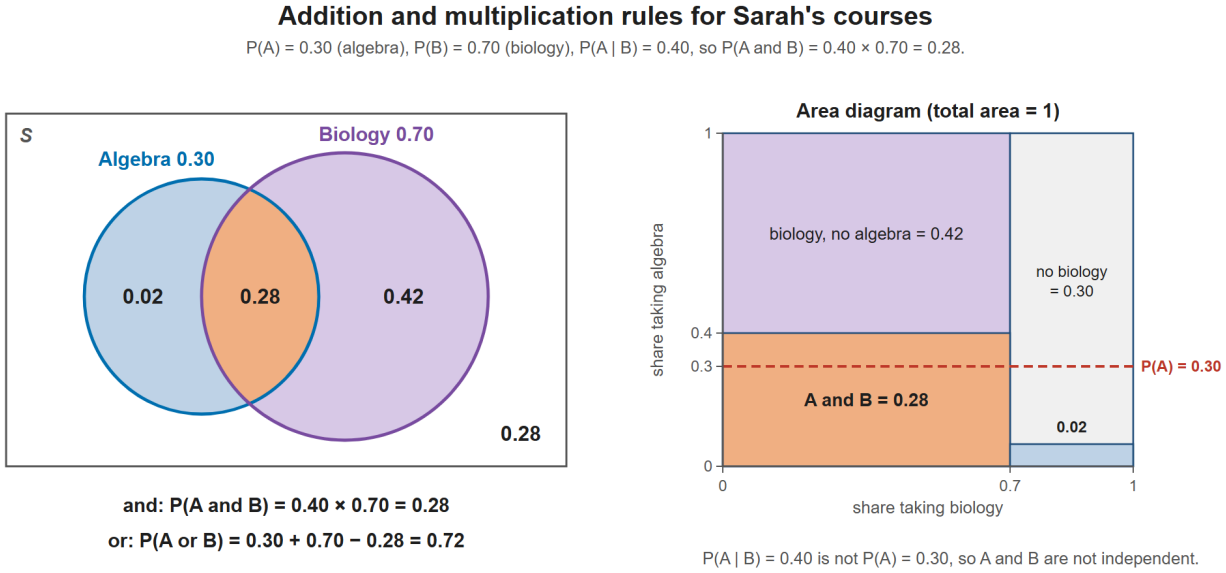
$$P(A \text{ and } B) = P(A \mid B)\,P(B) = 0.40 \times 0.70 = 0.28 = 28\%$$

(b) P(algebra *or* biology) - addition rule

$$P(A \text{ or } B) = 0.30 + 0.70 - 0.28 = 0.72 = 72\%$$

(c) Are the events independent? $P(A \mid B) = 0.40$ but $P(A) = 0.30$. They are not equal, so the events are **not independent**: taking biology makes algebra more likely.

(d) Are the events mutually exclusive? $P(A \text{ and } B) = 0.28 \neq 0$, so they are **not mutually exclusive**: she can take both.



Left: the four regions add up to 1 (0.02 + 0.28 + 0.42 + 0.28). Right: the same probabilities as areas. Within the biology column, algebra fills 40%, which is higher than the overall 30% (red line), so the events are not independent.

Key takeaways

- **Or** → **add** (and subtract the overlap). **And** → **multiply** (use a conditional probability unless the events are independent).
- Independent ≠ mutually exclusive:
  - independent: one doesn't affect the other, $P(A \cap B) = P(A)P(B)$
  - mutually exclusive: they can't both happen, $P(A \cap B) = 0$

Lesson 5.8 - Compound Probability of Independent Events

> **Video:** Compound Probability of Independent Events - Coins & 52 Playing Cards by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Rule

For **independent** events, multiply:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Example 1 - coins

(a) Two heads in two flips

$$P(HH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Check with the sample space $\{HH, HT, TH, TT\}$: 1 of 4 outcomes ✓

(b) Heads, tails, heads, in that order, in three flips

$$P(HTH) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Only 1 of the 8 outcomes ($HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$) matches.

(c) Two tails and one head, in any order The matching outcomes are HTT, THT, TTH:

$$P = \frac{3}{8}$$

> **Does order matter?** In (b) a specific sequence is required, so only one outcome counts. In (c) any arrangement counts, so add up all the matching outcomes.

A standard 52-card deck

Suit	Color	Cards
♦ Diamonds	Red	A, 2-10, J, Q, K (13)
♥ Hearts	Red	13
♠ Spades	Black	13
♣ Clubs	Black	13

- 4 suits × 13 = **52** cards; **26 red**, **26 black**; no jokers.
- Each rank (e.g. queen) appears **4** times, once per suit.

Example 2 - cards

(a) P(queen)

$$\frac{4}{52} = \frac{1}{13} \approx 0.0769 = 7.69\%$$

(b) P(black card numbered greater than 3 and at most 9) The numbers are 4, 5, 6, 7, 8, 9 (6 values), and there are two black suits (♠ and ♣):

$$\frac{6 \times 2}{52} = \frac{12}{52} = \frac{3}{13}$$

(c) P(red king first, then a diamond), *without* replacement

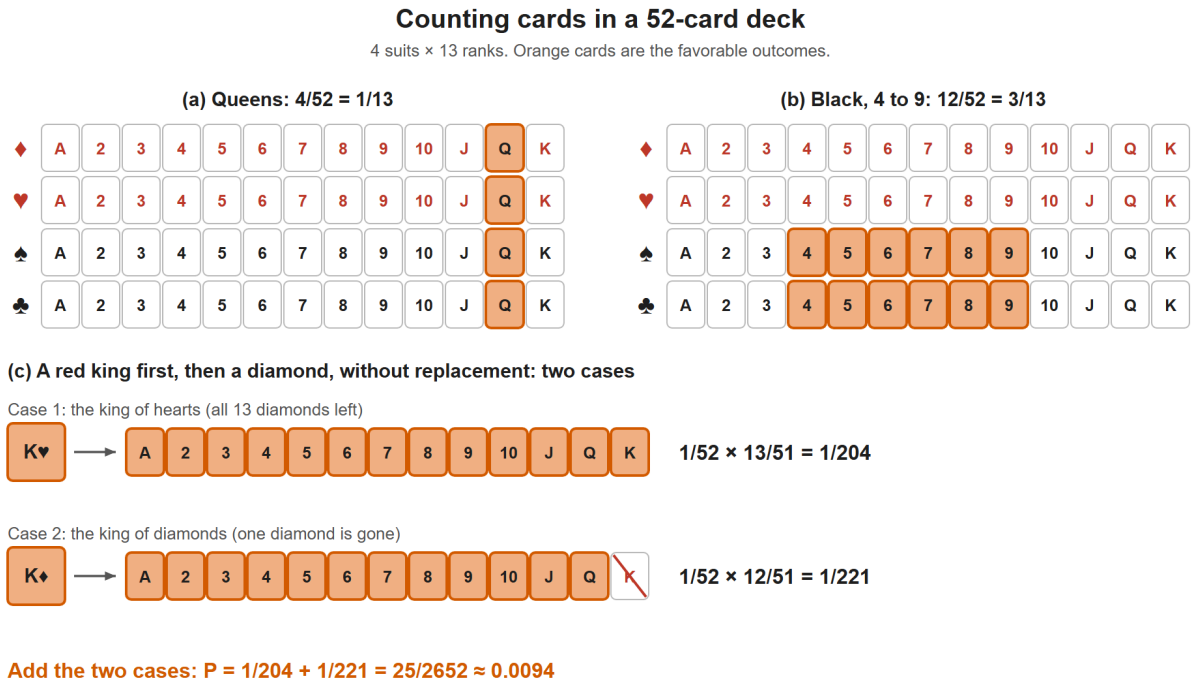
This is **dependent**: how many diamonds are left depends on *which* red king was drawn. Split it into two cases and add them (the cases are mutually exclusive):

Case	First card	Second card	Probability
1	King of ♥ ($\frac{1}{52}$)	any ♦; all 13 are still there ($\frac{13}{51}$)	$\frac{1}{52} \cdot \frac{13}{51} = \frac{1}{204}$
2	King of ♦ ($\frac{1}{52}$)	any ♦; only 12 are left ($\frac{12}{51}$)	$\frac{1}{52} \cdot \frac{12}{51} = \frac{1}{221}$

$$P = \frac{1}{204} + \frac{1}{221} = \frac{13}{2652} + \frac{12}{2652} = \frac{25}{2652} \approx 0.0094$$

(Common denominator: $2652 = 4 \cdot 3 \cdot 13 \cdot 17$.)

> Writing $P(\text{red king}) \times P(\text{diamond}) = \frac{2}{52} \times \frac{13}{51}$ would be **wrong**, because the second probability depends on which king was taken.



Counting favorable cards in the deck. For part (c) the number of diamonds left depends on which red king was taken first, so the two cases are worked out separately and added.

Key takeaways

- Independent events: multiply the probabilities. n fair coin flips in a specific order: $(1/2)^n$.
- "In any order" → count all the arrangements that work.
- Without replacement → dependent. If the second probability depends on *which* first outcome happened, split into cases and add.

Lesson 5.9 - How to Calculate the Expected Value

> **Video:** How To Calculate Expected Value by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Formula

The **expected value** is the long-run **average** result per trial:

$$E(X) = \sum x\,P(x) = x_1p_1 + x_2p_2 + \cdots + x_np_n$$

- x = the value of each outcome (gains positive, losses **negative**)
- p = its probability, as a **decimal**; all the p's add up to 1

Steps: make a table of *outcome* → *value* → *probability*, multiply each value by its probability, and add.

Expected total over n trials = $n \times E(X)$.

Example 1 - a casino game

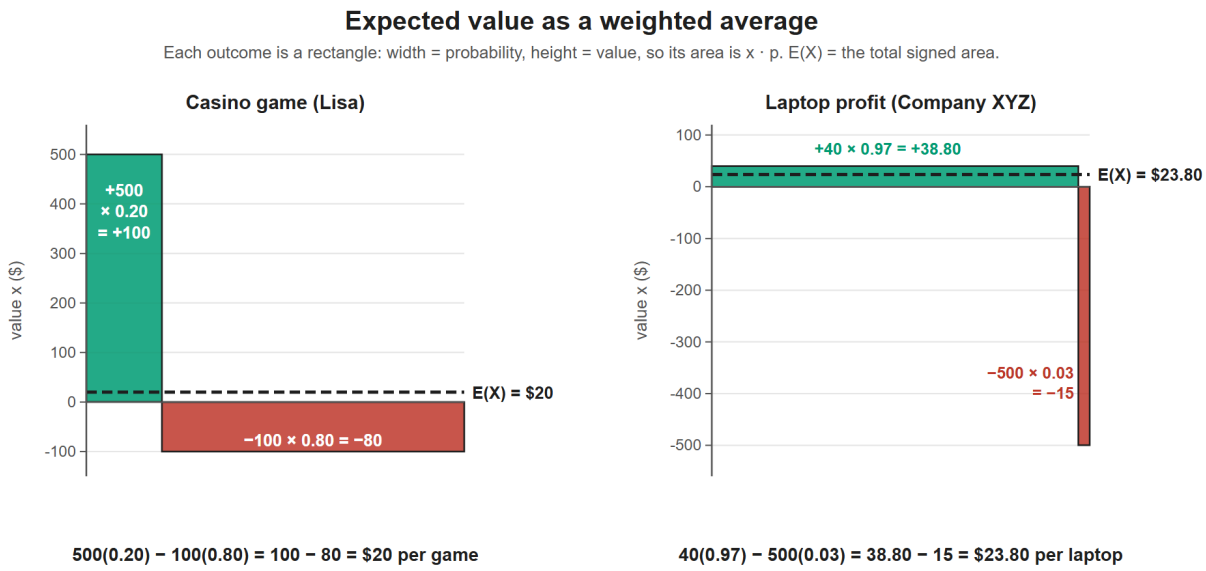
Lisa wins **$500** with probability 20%, and otherwise loses **$100**.

Outcome	Value x	Probability p	$x \cdot p$
Win	+500	0.20	+100
Lose	−100	0.80	−80
Total		**1.00**	**+20**

$$E(X) = 500(0.20) + (-100)(0.80) = 100 - 80 = \$20 \text{ per game}$$

- Over 10 games: $10 \times 20 = \$200$ expected.
- Over 100 games: $100 \times 20 = \$2{,}000$ expected.

(The probability of losing comes from the complement: $1 - 0.20 = 0.80$.)



Each outcome becomes a rectangle with area x times p. Positive areas (green) minus negative areas (red) give E(X). The dashed line at E(X) is the average height over the whole width.

Example 2 - profit per laptop

Company XYZ earns **$40** on each working laptop it sells and loses **$500** on each one returned as defective. **3 in 100** laptops are defective.

Outcome	Value x	Probability p	$x \cdot p$
Works	+40	0.97	+38.80
Defective	−500	0.03	−15.00
Total		**1.00**	**+23.80**

$$E(X) = 40(0.97) + (-500)(0.03) = 38.80 - 15.00 = \$23.80 \text{ per laptop}$$

Selling 100 laptops is expected to earn $100 \times 23.80 = \$2{,}380$.

Key takeaways

- $E(X) = \sum x\,p$: weight each outcome's value by its probability.
- Losses go in as **negative** values.
- The expected value is an **average over many trials**. A single game never pays exactly $20.
- Total expected = number of trials × expected value per trial.

Lesson 5.10 - Probability Tree Diagrams

> **Video:** Probability Tree Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

How a tree diagram works

- Each **branch** shows one possible outcome of a step, labeled with its probability.
- Branches leaving the same point **add up to 1**.
- **Along a path ("and")** → **multiply** the probabilities.
- **Combining paths ("or")** → **add** the path probabilities.
- All the final path probabilities add up to 1.

Problem 1 - with replacement (independent)

A bag holds **8 red** and **12 blue** marbles (20 total). Two marbles are drawn **with replacement**.

$P(R) = \frac{8}{20} = \frac{2}{5}$, $P(B) = \frac{12}{20} = \frac{3}{5}$. These stay the same on every draw.

```
                2/5   R  →  RR : 2/5 · 2/5 = 4/25
        R  <
  2/5 /         3/5   B  →  RB : 2/5 · 3/5 = 6/25
start
  3/5 \         2/5   R  →  BR : 3/5 · 2/5 = 6/25
        B  <
                3/5   B  →  BB : 3/5 · 3/5 = 9/25
```

Check: $4 + 6 + 6 + 9 = 25$, so the total is $25/25 = 1$ ✓

(a) P(two blue) $= \frac{9}{25} = 0.36 = 36\%$

(b) P(at least one red): paths RR, RB, BR

$$\frac{4 + 6 + 6}{25} = \frac{16}{25} = 0.64$$

Complement shortcut: "at least one red" = "not both blue":

$$1 - P(BB) = 1 - 0.36 = 0.64 \checkmark$$

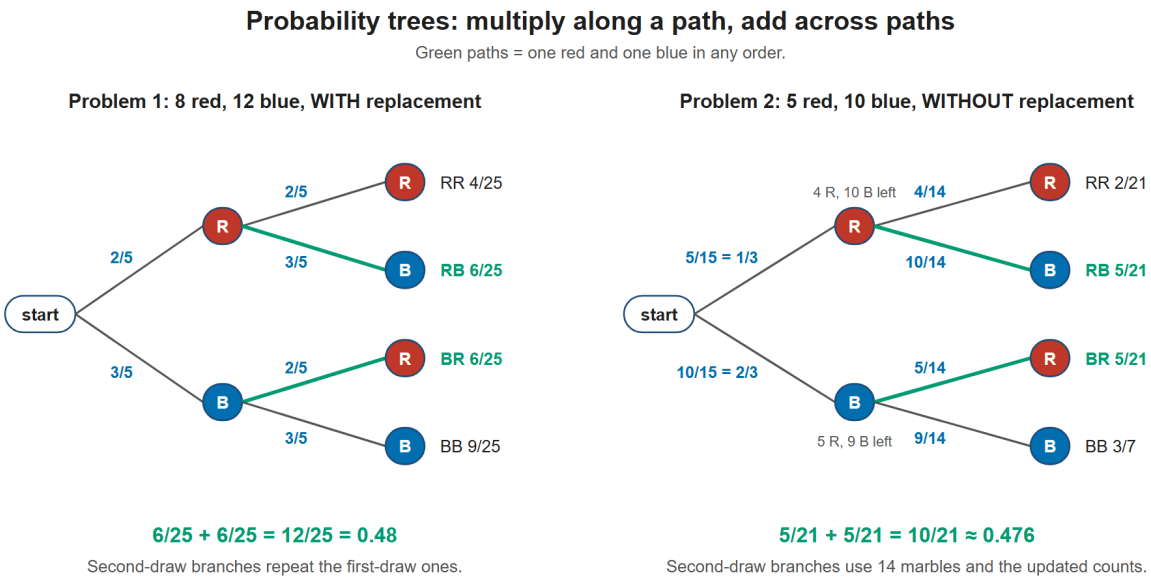
(c) P(red first, then blue): only RB

$$\frac{6}{25} = 0.24 = 24\%$$

(d) P(one red and one blue), in any order: RB or BR

$$\frac{6}{25} + \frac{6}{25} = \frac{12}{25} = 0.48 = 48\%$$

> (c) and (d) differ: "red **then** blue" fixes the order, while "a red **and** a blue" allows either order.



Multiply along each path to get its probability, then add the paths that match the question. Without replacement, the second-draw branches change depending on the first draw.

Problem 2 - without replacement (dependent)

A bag holds **5 red** and **10 blue** marbles (15 total). Sally draws two **without replacement**, so the second-draw probabilities depend on the first draw.

```
                4/14 = 2/7   R   →  RR : 1/3 · 2/7   = 2/21
        R   <                       (4 R, 10 B left)
  1/3   /       10/14 = 5/7  B   →  RB : 1/3 · 5/7   = 5/21
start
  2/3   \       5/14         R   →  BR : 2/3 · 5/14  = 5/21
        B   <                       (5 R, 9 B left)
                9/14         B   →  BB : 2/3 · 9/14  = 3/7
```

(a) P(two red)

$$\frac{1}{3} \cdot \frac{2}{7} = \frac{2}{21} \approx 0.095 = 9.5\%$$

(b) P(one red and one blue), in any order

$$P(RB) + P(BR) = \frac{5}{21} + \frac{5}{21} = \frac{10}{21} \approx 0.476 = 47.6\%$$

Problem 3 - three draws with replacement

3 blue, **7 red** (10 total). Megan draws 3 **with replacement**: $P(R) = 0.7$, $P(B) = 0.3$ every time.

There are $2^3 = 8$ outcomes: RRR, RRB, RBR, RBB, BRR, BRB, BBR, BBB.

(a) P(at least one red): every outcome except BBB

$$1 - P(BBB) = 1 - (0.3)^3 = 1 - 0.027 = 0.973 = 97.3\%$$

(b) P(at most two blue): 0, 1 or 2 blue, again everything except BBB

$$1 - 0.027 = 0.973$$

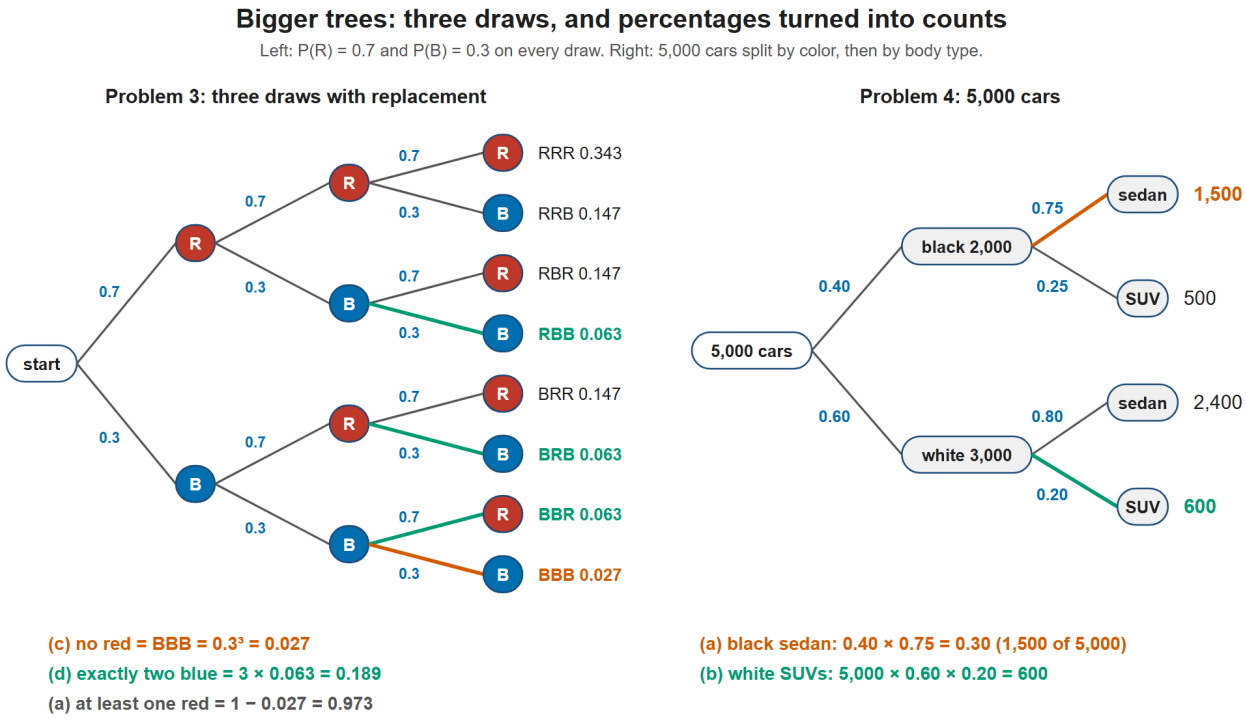
"At least one red" and "at most two blue" describe the **same** event.

(c) P(no red) = P(BBB)

$$(0.3)^3 = 0.027 = 2.7\%$$

(d) P(exactly two blue): RBB, BRB, BBR

$$3 \times (0.7)(0.3)(0.3) = 3 \times 0.063 = 0.189 = 18.9\%$$



Left: the 8 outcomes of three draws. Exactly two blue is three separate paths. Right: multiplying the counts by the branch percentages turns the tree into counts.

Problem 4 - percentages on a tree

There are **5,000** cars in a neighborhood. **40%** are black and **60%** are white. **75%** of the black cars are sedans (the rest SUVs). **20%** of the white cars are SUVs (the rest sedans).

```
                        0.75  sedan   1,500
         black  2,000 <
  0.40 /                0.25  SUV       500
5,000
  0.60 \                0.80  sedan   2,400
         white  3,000 <
                        0.20  SUV       600
```

(a) P(black sedan)

$$\frac{1500}{5000} = 0.30, \quad \text{or along the path: } 0.40 \times 0.75 = 0.30 = 30\%$$

(b) Number of white SUVs

$$5000 \times 0.60 \times 0.20 = 600$$

Key takeaways

- Multiply **along** a branch; add **across** branches.
- **With replacement:** the branch probabilities repeat. **Without replacement:** update both the counts and the total after each draw.
- "At least one ..." is usually easiest as $1 - P(\text{none})$.
- Watch for "in that order" (one path) versus "in any order" (several paths).

Lesson 5.11 - Bayes' Theorem of Probability with Tree Diagrams

> **Video:** Bayes' Theorem of Probability With Tree Diagrams & Venn Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Deriving Bayes' theorem

Start from two conditional probabilities:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Both give the same joint probability:

$$P(A \mid B)\,P(B) = P(A \cap B) = P(B \mid A)\,P(A)$$

Divide by $P(B)$:

$$\boxed{P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}}$$

Bayes' theorem gives a conditional probability when you know the **reverse** one, $P(B \mid A)$.

Example 1 - slips of paper (checking the formula)

A bottle holds slips numbered 1-9, each equally likely to be drawn. $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8, 9\}$, $A \cap B = \{4, 5\}$

Venn regions: A only = {1, 2, 3}; both = {4, 5}; B only = {6, 7, 8, 9}.

- $P(A) = \frac{5}{9}$, $P(B) = \frac{6}{9}$
- $P(B \mid A)$: of A's 5 numbers, 2 are in B, so $\frac{2}{5}$

Using Bayes:

$$P(A \mid B) = \frac{\frac{2}{5} \cdot \frac{5}{9}}{\frac{6}{9}} = \frac{2/9}{6/9} = \frac{2}{6} = \frac{1}{3}$$

Check with the direct formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/9}{6/9} = \frac{1}{3} \checkmark$$

Example 2 - medical screening

- 12% of men develop prostate cancer: $P(C) = 0.12$
- With cancer, the test is positive 95% of the time: $P(+ \mid C) = 0.95$
- Without cancer, there is a 6% false-positive rate: $P(+ \mid C') = 0.06$

Find $P(C \mid +)$, the chance of having cancer given a positive test.

Step 1 - tree diagram

```
                0.95  positive   →  0.12 × 0.95 = 0.1140
        C  <
 0.12 /         0.05  negative   →  0.12 × 0.05 = 0.0060
start
 0.88 \         0.06  positive   →  0.88 × 0.06 = 0.0528
        C' <
                0.94  negative   →  0.88 × 0.94 = 0.8272
```

Step 2 - total probability of a positive test

A positive result happens through either branch, so add them:

$$P(+) = P(C)P(+ \mid C) + P(C')P(+ \mid C') = 0.1140 + 0.0528 = 0.1668$$

Step 3 - Bayes

$$P(C \mid +) = \frac{P(+ \mid C)\,P(C)}{P(+)} = \frac{0.95 \times 0.12}{0.1668} = \frac{0.114}{0.1668} \approx 0.683$$

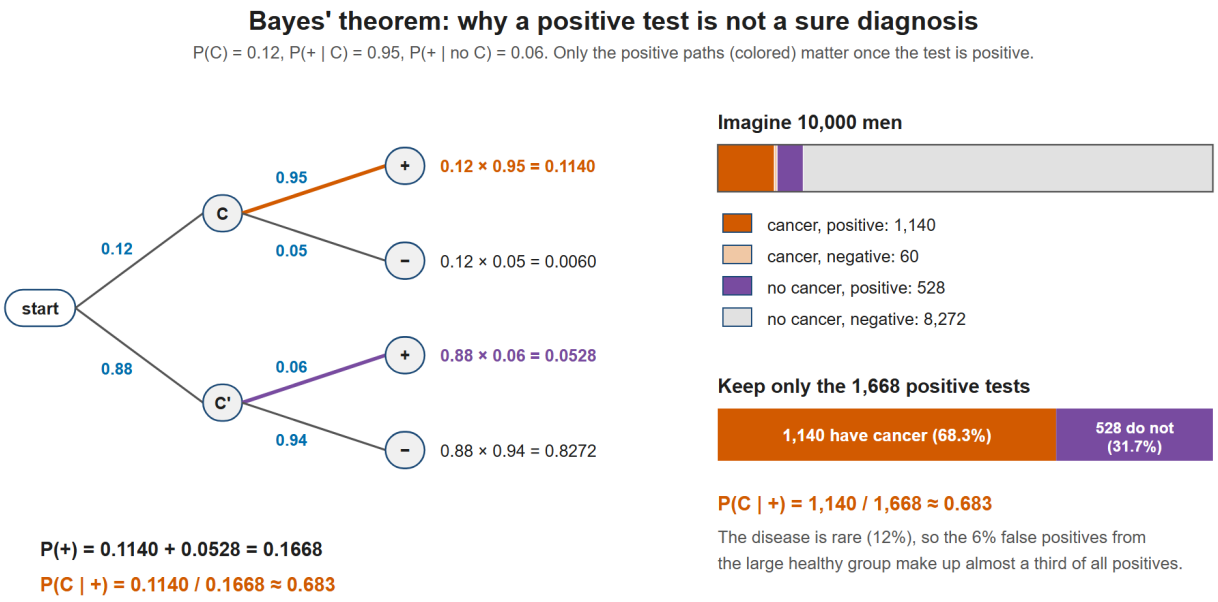
A man with a positive test has about a **68.3%** chance of having cancer, so about **31.7%** of positive results are false alarms ($1 - 0.683$).

Same answer with counts (imagine 10,000 men)

	Positive	Negative	Total
Cancer (12%)	1,140	60	1,200
No cancer (88%)	528	8,272	8,800
Total	**1,668**	8,332	10,000

$$P(C \mid +) = \frac{1140}{1140 + 528} = \frac{1140}{1668} \approx 0.683 \checkmark$$

The choice of 10,000 doesn't matter, because only the ratio counts.



Out of 10,000 men, 1,668 test positive, but only 1,140 of them have cancer. Restricting to the positive tests gives P(C | +) = 1,140 / 1,668 ≈ 0.683.

Key takeaways

- $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$, which turns a known conditional probability around.
- The denominator often has to be built from a tree: $P(B) = P(B \mid A)P(A) + P(B \mid A')P(A')$.
- A positive test does **not** guarantee the condition. Rare conditions plus false positives can produce many false alarms.
- Converting to counts (e.g. 10,000 people) is an easy way to check the answer.